Product Requirements Document

AI Try-On — Online Jewellery Shopping App

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1. Goal

Let users see jewellery on their own face and neck before they buy, so the purchase decision is based on fit and personal look rather than a catalogue photo alone.

2. Vision / Big Picture

AI Try-On becomes the default way people evaluate fine jewellery online — the visual-confidence layer that closes the gap between browsing and buying.

- Reduce reliance on in-store trial for first-time high-value purchases.
- Make the app the reference point customers reopen before every occasion-based purchase, building habitual, non-transactional engagement.
- Differentiate against marketplaces that only show model photography.

3. Background

The two founding pain points are catalogue mistrust and outfit-matching uncertainty. Neither is solved by better photography alone — polishing catalogue shots does not tell a user how a piece will sit on their own neck, skin tone, or outfit.

Assumption: *We do not have first-party usage data for this exact feature yet, since it doesn't exist in the app today. The figures below are directional, drawn from category benchmarks for AR/virtual try-on in beauty and eyewear e-commerce (where try-on is established), adjusted down for jewellery's higher price point and lower purchase frequency.*

Signal	Value	Why it matters
Return rate on jewellery purchased sight-unseen (industry estimate)	18–22%	A meaningful share of returns cite "did not look as expected" rather than defect or damage.
Session-to-cart conversion on PDP, current app	~2.1%	Baseline we expect Try-On to move, not create from zero.
Support queries referencing "how will this look on me"	~9% of pre-sales chat volume	Direct evidence users are asking the question manually; we are productising an existing behaviour.

These are working estimates for planning purposes, to be replaced with actual data once analytics instrumentation (Section 9) is live and we have 4–6 weeks of Phase 1 data.

Why pursue this now, specifically: camera-based face/neck landmark tracking has become reliable enough on mid-range phones (not just flagship) to make real-time overlay viable without a native app rebuild — this was not true 2–3 years ago and is the main reason this wasn't built earlier.

4. North Star Metric

Metric	Baseline	Target	Timeframe
% of PDP sessions that use Try-On before Add to Cart	0% (new)	25%	6 months post full launch

Why this metric, not a revenue number directly: Try-On adoption is a leading indicator we can move quickly and attribute cleanly. Revenue and return-rate impact (Section 11) are lagging outcomes we track alongside it, but using them as the North Star would take 2–3 purchase cycles to show signal and would be confounded by pricing, seasonality, and marketing spend.

5. User Personas

Persona	Goal	Pain Points (prioritised)
The Occasion Shopper(buying for wedding/festival, high basket value, 1–2 purchases/year)	Wants certainty before spending ₹50K+ on a single piece	1. Doesn't trust catalogue images — lighting and model skin tone differ from her own 2. Can't tell if a piece suits her outfit until it physically arrives 3. Reluctant to return high-value items — return friction is high and emotionally uncomfortable
The Everyday Gold Buyer(smaller, frequent purchases, price-sensitive)	Wants a quick, low-effort way to check "does this suit me" without overthinking	1. Won't spend more than 30–45 seconds evaluating a sub-₹20K piece 2. Low tolerance for a slow or fiddly Try-On flow — will abandon if it feels like a chore
The Gifter(buying for someone else, e.g. spouse/parent)	Wants confidence the piece will suit the recipient, without the recipient present	1. Cannot try the piece on the actual recipient 2. Relies on guessing skin tone/face shape from memory or an old photo

Assumption: *The Gifter persona is real and common in this category (gifting is a large share of Indian jewellery purchases), but Try-On v1 does not solve for her directly — she can upload a photo of the recipient with consent, but we are not building a "try on someone else's face from a saved photo" flow as a first-class feature. Flagged explicitly in Section 13 as a decision Design and Legal should weigh in on given consent implications.*

6. Pain Points → Features Mapping

Pain Points	Feature
Users don't trust catalogue images because lighting, editing and model photos make products appear different	Live camera Try-On with neutral, corrected lighting normalization; Comparison View (before/after, side-by-side)
Users don't know whether jewellery matches their outfit	Photo-upload Try-On (so a user can try against a photo of themselves in the actual outfit); Saved Looks to revisit before deciding
High-value returns due to "didn't look as expected" (inferred, Section 3)	Match confidence indicator on Result Screen; Try-On badge carried into Cart so the decision context isn't lost
Slow, effortful evaluation for lower-value/frequent purchases	Sub-5-second processing target (Section 9); one-tap Retry

7. User Journey Flow

Written as the primary happy path, with explicit branches for the two entry points (camera vs. upload) and failure states called out inline rather than bolted on at the end — Engineering needs to build these branches from day one, not retrofit them.

7.1 Entry Point

- Primary: "Try It On" CTA on PDP, placed directly beneath the price block (highest-visibility position without competing with Add to Cart).
- Secondary: "Try-On" tab in bottom navigation → opens a picker of recently viewed / trending pieces (for users who want to browse via Try-On rather than the catalogue first).

7.2 Permissions

- On first tap of "Try It On," request camera permission with a pre-permission explainer screen (not the raw OS dialog cold) — explains why we need it and that no photo is uploaded without explicit save action.
- If camera permission is denied: fall back to the Upload flow automatically, with a persistent "Enable camera in Settings" link. Do not dead-end the user.
- If Upload is used: request photo-library permission with the same explainer pattern.

7.3 Camera Flow

- User is shown a live front-camera feed with a face/neck alignment guide overlay.
- Real-time landmark detection tracks position; guide text updates live ("Align chin within the guide," "Move a little further back").
- Capture is manual (shutter tap), not auto-triggered — auto-capture in earlier internal reviews was flagged as disorienting when lighting or angle is marginal; manual capture also gives the user a clear moment of control.

7.4 Gallery Upload (alternate path)

- User selects an existing photo. We require a face to be detectable in the photo before proceeding — if no face is detected, show an inline error with example guidance ("Choose a clear, front-facing photo") rather than a generic failure.

7.5 AI Processing

- Processing screen shown with progress indicator and target completion time communicated ("Usually takes 3–5 seconds") so the wait feels bounded.
- Backend performs face/neck landmark mapping, lighting/skin-tone normalization, and jewellery asset overlay (Section 9 has the technical detail).

7.6 Result Screen

- Rendered overlay shown full-screen with product name, match confidence, and four actions: Compare, Retry Photo, Try Another (piece), Save Look.

7.7 Retry

- Retry Photo returns to capture/upload with the previous frame discarded — not silently reused, to avoid the user assuming a stale result is being re-rendered.
- If two consecutive processing attempts fail (Section 7.9), Retry is temporarily replaced with "Report an issue" plus a manual fallback to catalogue images, so the user isn't stuck retrying a broken state indefinitely.

7.8 Save / Share / Add to Cart

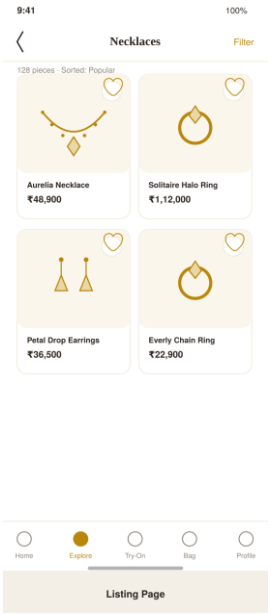
- Save Look stores the rendered image + product reference for 30 days, visible in a "Saved Looks" section.
- Share exports a watermarked image (brand + product name) to the native share sheet — watermarking exists so shared images double as organic marketing, and so a shared image can't be mistaken for a "real" worn photo out of context.
- Add to Cart from the Result Screen carries a "Tried On" tag into the cart line item, preserved through Checkout so downstream teams (CS, merchandising) can see Try-On-influenced orders.

7.9 Failure States

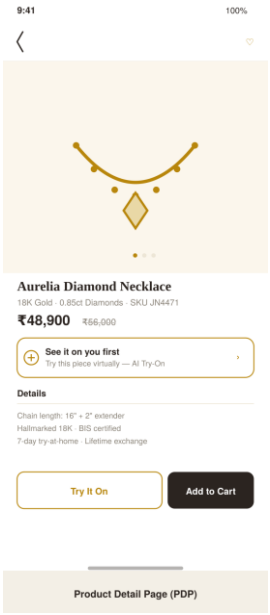
Failure	Handling
No face detected in frame/photo	Inline guidance + retry; does not consume a processing attempt against rate limits
Poor lighting (under/over-exposed)	Warn before capture where detectable in real time; if only caught post-capture, surface as a specific error, not a generic "something went wrong"
Processing timeout (>10s)	Auto-cancel, apologise, offer Retry; log for the AI Accuracy dashboard (Section 11)
Jewellery asset unavailable for this SKU (3D/overlay model not yet generated)	"Try-On not yet available for this piece" state on PDP — CTA is hidden/disabled rather than shown and failing, so we never promise something we can't deliver
Device/browser unsupported (Section 9)	Try-On entry point is hidden at the client level rather than shown and failing — detected via capability check on page load
Network drop mid-processing	Local retry with cached frame for up to 30s before requiring a fresh capture

8. Mockups

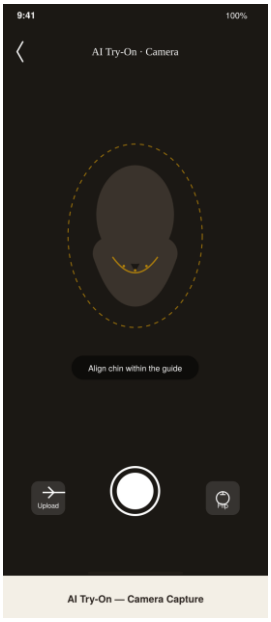
Full-colour mobile mockups below, styled consistent with premium jewellery retail conventions (ivory/white base, gold accents, serif product typography). These are directional for Design to iterate on, not final visual specs.



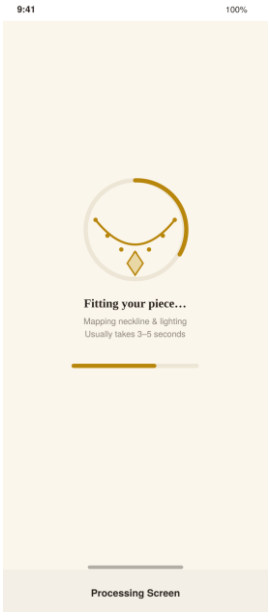
Listing Page — Try-On not yet in context; intent here is still browsing.



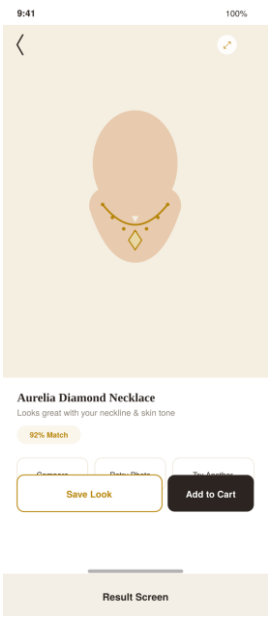
"Try It On" as a secondary CTA beneath price, distinct from Add to Cart.



Camera — dark chrome keeps focus on the live feed; manual shutter.



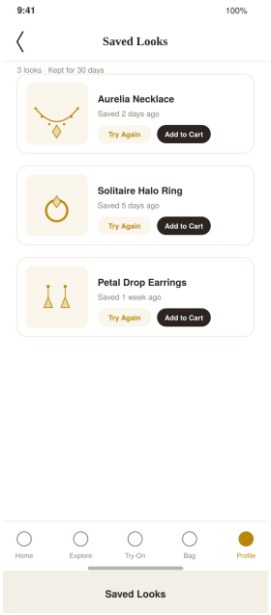
Processing — bounded wait time is communicated explicitly.



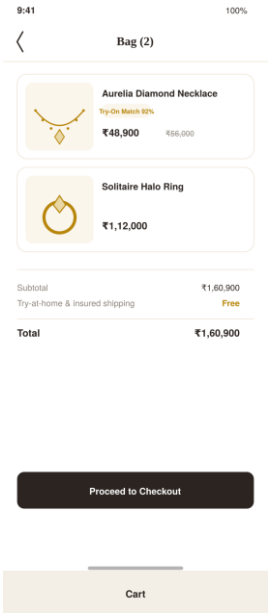
Result Screen — match confidence and four clear next actions.



Comparison View — before/after, plus previously tried pieces.



Saved Looks — 30-day retention, quick Try Again / Add to Cart.



Cart — "Tried On" context (match %) carried through to Cart.

9. Technical Requirements

9.1 Frontend

- Camera access via native device camera APIs (WebRTC getUserMedia for web/PWA context, native camera APIs for iOS/Android app).
- Real-time face/neck landmark overlay rendered client-side for the alignment guide (lightweight model, on-device) — the alignment guide must not depend on a server round-trip, or it will feel laggy.
- Final try-on render composited server-side, not client-side, for consistent quality across device tiers and to protect the overlay model IP.
- Graceful capability detection: front camera availability, WebGL/GPU support for rendering, and minimum OS version, checked on PDP load before showing the Try-On CTA.

9.2 Backend

- New Try-On Service, exact endpoint structure owned by Engineering. At a minimum it needs to support: creating a try-on session; accepting a submitted frame/photo for processing; returning a result (rendered overlay + match confidence + product reference); and persisting a result against a user account on explicit Save.
- Data payload requirements: session creation needs user id, SKU id, and entry point (camera vs. upload); frame submission needs the image payload plus device/capability metadata (for the AI Accuracy dashboard in Section 11); the result payload needs the rendered image reference, a match-confidence score, and a processing-duration figure so latency (Section 9.6) is measurable end to end.
- Async processing queue (capture submission returns immediately with a session reference; result delivered via short-poll or WebSocket, Engineering's call) so the client isn't holding an open HTTP connection for the full 3–5s.
- Rate limiting per user (e.g., 30 processing attempts/hour) to control inference cost and abuse.

9.3 AI / Computer Vision

- Face and neck landmark detection model (68+ point facial landmarks, neck contour estimation) to anchor jewellery placement.
- Per-SKU 3D/overlay asset required for each try-on-enabled product — this is a content production dependency, not just an engineering one; Catalogue/Merchandising must generate and QA these assets ahead of enabling Try-On per SKU. This is the single biggest execution risk in this PRD: producing high-quality overlay assets for the full necklace catalogue is a slower, more expensive pipeline than any of the engineering work above, and is likely to lag the rest of Phase 1 if not started in parallel from week one.
- Lighting and skin-tone normalization applied to the overlay (not the user's underlying photo) so the jewellery reads accurately regardless of ambient light, without altering how the user looks otherwise.
- Metal and stone rendering must simulate specular reflection and subsurface scattering, not a flat texture map — a flat-shaded overlay is the fastest way to undermine the trust this feature exists to build, since gold and diamonds are defined visually by how they catch light. This needs to be validated per skin tone and per ambient lighting condition, not just against a single reference photo.
- Model versioning: overlay accuracy will improve iteratively; we need A/B capability between model versions and a rollback path if a new version regresses accuracy metrics (Section 11).

9.4 Image Processing & Storage

- Captured frames are processed in-memory and are not written to persistent storage by default — they exist only for the duration of the active session and are discarded immediately after the result is returned, unless the user explicitly taps Save Look.
- Saved Looks (explicit opt-in only) stored for 30 days, then auto-purged, unless the user re-saves.
- Images stored encrypted at rest; access scoped to the owning user account.

Assumption: *Under India's Digital Personal Data Protection (DPDP) Act, facial imagery is sensitive personal data and processing it requires explicit, specific consent — general ToS acceptance is not sufficient. We assume Legal will require: (a) a standalone consent step before any camera/photo processing begins, separate from account-level ToS; (b) no server-*

side persistence of a frame unless Save Look is explicitly tapped; and (c) a clear, user-facing deletion path that cascades to backend storage. This is flagged as a blocking open question in Section 13, not a settled decision — the exact consent language and retention posture need Legal sign-off before Phase 1 ships.

9.5 Analytics

- Event instrumentation at every journey step in Section 7 (entry, permission grant/deny, capture, processing start/end/fail, result view, retry, save, share, add-to-cart-from-tryon) — this is what makes Sections 4 and 11 measurable from day one rather than retrofitted.
- Processing latency and failure-reason logged per session for the AI Accuracy dashboard.

9.6 Latency Expectations

Step	Target
Camera feed to alignment guide active	< 500ms
Capture to processing start	< 300ms
Processing (server render)	3–5s target, 10s hard timeout
Result screen render	< 500ms after processing completes

9.7 Permissions & Privacy

- Camera and photo-library permissions requested only at point of use, with explainer screens (Section 7.2), never pre-emptively on app install.
- Clear in-flow disclosure of what is stored (only on explicit Save) versus what is processed transiently and discarded.
- User can delete any Saved Look and revoke camera permission at any time from Settings; deletion must cascade to backend storage, not just hide client-side.

9.8 Supported Devices

- iOS 15+ and Android 10+ (covers the large majority of active devices per typical app analytics; exact cutoff to be confirmed against our actual device distribution before final commitment).
- Minimum 3GB RAM devices for acceptable real-time overlay performance — below this, we fall back to upload-and-process (no live camera guide) rather than a degraded live experience.

9.9 Unsupported Scenarios

- Very low light (below a defined lux threshold detectable via frame brightness analysis) — user is warned before capture rather than receiving a poor-quality result.
- Side-profile-only photos for necklace/earring try-on (front-facing required for reliable landmark detection).
- Users wearing existing large neckwear/high collars that occlude the neck region for necklace try-on — flagged as a known accuracy limitation, not silently mis-rendered.

10. Release Plan

Feature	Priority	Phase	Rationale
Camera Try-On, necklaces only, top 50 bestselling SKUs	P0	Phase 1 (MVP)	Necklaces chosen first — simplest anchor geometry (neck contour) vs. rings (hand pose is harder to track reliably) or earrings (occlusion from hair). Capped at 50 SKUs

			deliberately: overlay asset production (Section 9.3) is the real bottleneck, and launching Try-On broadly with the CTA missing or failing on 90%+ of the catalogue would damage trust faster than it builds it.
Photo Upload Try-On	P0	Phase 1 (MVP)	Directly solves the outfit-matching pain point (Section 6); low incremental engineering cost once camera pipeline exists
Result Screen + Add to Cart integration	P0	Phase 1 (MVP)	Without this the feature doesn't close the loop into a business outcome
Expand to full necklace catalogue	P1	Phase 2	Gated on asset-production throughput, not engineering readiness — Merchandising needs a validated per-SKU turnaround time from Phase 1 before we commit a date
Comparison View	P1	Phase 2	High value but not blocking for MVP validation; needs Saved state to exist first
Saved Looks	P1	Phase 2	Depends on storage/privacy consent flow (Section 9.4) being finalised
Try-On for rings	P1	Phase 2	Requires hand-pose detection — materially different CV problem from face/neck; separate model investment
Try-On for earrings	P2	Phase 3	Occlusion (hair, ears partially hidden) makes this the hardest of the three categories
Share to social	P2	Phase 3	Marketing upside but not core to the purchase-decision problem this PRD is scoped to solve

Phase 1 is deliberately scoped to necklaces only, on a curated 50-SKU subset, so we can validate the North Star metric and pressure-test the asset-production pipeline (Section 9.3) before committing to the full catalogue in Phase 2.

11. Metrics

Category	Metric	Notes
Activation	% of PDP-viewing users who tap "Try It On" at least once	Top of funnel for the whole feature
Engagement	Try-Ons per session; % using Retry; % reaching Result Screen	Retry rate signals friction, not just interest
Conversion	Add-to-Cart rate from Result Screen vs. PDP baseline (Section 3)	Primary business validation of the feature's ROI

Business	Revenue and order count on Try-On-tagged orders; AOV comparison vs. non-Try-On orders	Watch for AOV lift, not just conversion lift
Trust	Return rate on Try-On-influenced orders vs. baseline (Section 3)	This is the metric that ultimately justifies the investment — needs 6–8 weeks post-purchase to mature per cohort
AI Accuracy	Processing success rate; average latency; user-reported "didn't look right" rate	Self-reported dissatisfaction post-purchase is a proxy for overlay accuracy we can't measure directly
Drop-off	% abandoning at permission prompt; % abandoning between capture and result; % abandoning after result without action	Each is a different fix — permission copy, processing speed, or result-screen CTA clarity respectively

12. GTM Strategy

- What: Launch as a visible, named feature ("AI Try-On") rather than a quiet backend addition — the value only lands if users know to look for it.
- Who: Necklace-category shoppers first (matches Phase 1 scope); prioritise users who have viewed 3+ necklace PDPs in the last 30 days without purchasing, as the segment most likely to have hit the trust/fit uncertainty this feature targets.
- Where: In-app PDP placement (primary) + push notification to the above segment + homepage banner for the first two weeks post-launch.
- When: Soft launch to 10% of eligible traffic for one week (monitor latency and failure rates under real load) before ramping to 100%.

13. Open Questions

Question	Owner
What exact consent language and retention posture does the DPDP Act require for facial imagery — does our in-memory-only / opt-in-Save-only approach (Section 9.4) satisfy it, or does Legal need something stricter (e.g., time-bound consent renewal)?	Legal
What is our actual device/OS distribution — does the proposed iOS 15+/Android 10+ cutoff exclude a meaningful user segment?	Engineering
What is a realistic per-SKU turnaround time for overlay asset production, and does that support expanding past the initial 50 SKUs on the Phase 2 timeline, or does it push Phase 2 out?	Design/Category
Should the Gifter persona's "try on someone else's photo" use case be explicitly supported with consent language, explicitly blocked, or left ambiguous for Phase 1?	Product/Legal
What is our fallback UX if a device fails the capability check but the user is on a high-value cart — do we ever offer a manual/human alternative (e.g., video call with a stylist)?	Product
Is 30 days the right Saved Looks retention window, or should it match typical purchase-decision cycles more closely (which may be longer for high-value pieces)?	Product/Data

14. Future Scope

Feature	Purpose	Priority	Timeframe
Multi-piece styling (necklace + earrings together)	Reflects real purchase behaviour — jewellery is often bought/worn as a set, not a single piece	P1	Post Phase 3
Stylist-curated "Complete the Look" suggestions surfaced on Result Screen	Cross-sell built on top of Try-On data, not a separate recommendation system	P2	Post Phase 3
In-store kiosk version of Try-On	Extends the same trust-building mechanic to offline, for stores that stock less than the full online catalogue	P2	6+ months out
Video Try-On (short clip instead of static frame) to see movement/shine	Static overlay can't show how a piece catches light when the wearer moves — a known limitation surfaced in Section 9.9	P2	6+ months out